

Local AI-Assisted Smart Traffic Signal Control for Emergency Vehicle Priority

SHAKUNTHALA G

Department of Data Science and Engineering
Bellary Institute of Technology And Management
Bellary, Karnataka, India
shakunthalag15072007@gmail.com

Abstract—Traditional traffic signals generally operate using fixed timing plans and therefore have limited ability to respond to rapid changes in traffic conditions. Adaptive traffic signal control and machine-vision-based approaches can estimate traffic conditions and dynamically modify signal operation. However, emergency vehicles such as ambulances require additional priority because delay at signalized intersections can affect response time.

This paper proposes a conceptual local-AI-assisted traffic signal architecture that combines camera-based traffic observation, local processing, emergency vehicle detection, and coordinated signal priority. The proposed architecture processes traffic information close to the intersection, thereby reducing dependence on continuous cloud communication. A research gap analysis shows that adaptive traffic control, machine vision, and emergency vehicle preemption have each been studied extensively, while their integration into a resource-conscious local architecture with coordinated emergency priority and recovery of non-priority traffic remains a useful area for further investigation.

Index Terms—Smart traffic lights, local AI, edge computing, machine vision, emergency vehicle, traffic signal control, emergency vehicle preemption

I. INTRODUCTION

Urban traffic congestion is a major challenge for intelligent transportation systems. Conventional traffic signals commonly use predetermined timing plans. Although fixed-time control is simple to implement, it cannot always respond effectively to rapid changes in traffic demand.

Artificial intelligence and machine vision have therefore been investigated for adaptive traffic signal control. A review by Rasheed *et al.* discusses the use of deep reinforcement learning for traffic signal control and identifies several research directions for intelligent traffic management [1].

Machine-vision-based traffic signal control has also been investigated. Gorodokin *et al.* developed an adaptive traffic-light control approach using machine vision and vehicle-flow information to modify signal timing according to traffic conditions [2].

However, normal traffic optimization is not sufficient for emergency situations. Emergency vehicle preemption gives priority to emergency vehicles at signalized intersections. Previous studies have investigated different strategies for emergency vehicle preemption and the effect of these strategies on other traffic [3].

Therefore, this paper considers a local-AI-assisted architecture that combines traffic observation, emergency vehicle detection, and traffic-signal control.

The main objectives of this work are:

- To describe a local AI architecture for adaptive traffic signal control.
- To incorporate emergency vehicle detection into the traffic-control process.
- To reduce dependence on continuous cloud communication.
- To coordinate signal priority for an emergency vehicle.
- To consider the recovery of normal traffic after emergency preemption.

II. RELATED WORK

A. AI-Based Traffic Signal Control

Deep reinforcement learning has become an important research direction for traffic signal control. Rasheed *et al.* reviewed reinforcement-learning-based traffic signal control and discussed architectures, simulation environments, complexity, and open research issues [1].

These approaches demonstrate that intelligent controllers can adapt signal decisions to changing traffic conditions. However, AI-based control can require considerable computation and suitable training environments.

B. Machine Vision for Adaptive Traffic Signals

Machine vision provides a way to obtain traffic information directly from cameras. Gorodokin *et al.* investigated machine-vision-based adaptive traffic light control using vehicle-flow information and a neural-network-based detection approach [2].

Such systems can estimate vehicle numbers, queue lengths, and traffic characteristics. This makes camera-based sensing useful for adaptive traffic control.

C. Emergency Vehicle Preemption

Emergency vehicle signal preemption has been studied for many years. Qin and Khan investigated control strategies for transitioning traffic signals to emergency vehicle preemption while attempting to reduce the effect on general traffic [3].

An IEEE study also investigated an emergency-vehicle-centered traffic signal control approach designed to reduce

emergency vehicle delay while limiting negative effects on conflicting traffic [4].

More recent work has considered queue spillbacks and recovery of normal traffic along emergency-vehicle routes. Chen *et al.* proposed an emergency vehicle traffic signal preemption system that considers queue spillbacks and the effects of preemption on non-priority traffic [5].

III. RESEARCH GAP

The literature demonstrates that three important areas have already received significant attention:

- 1) Adaptive traffic signal control using AI.
- 2) Machine-vision-based traffic estimation.
- 3) Emergency vehicle signal preemption.

Therefore, the research gap should not be stated as the absence of emergency vehicle preemption.

Instead, this study identifies an integration-oriented gap. Existing studies demonstrate individual or combined solutions for adaptive control and emergency preemption, but there is still scope for a practical architecture that combines:

- local camera-based traffic analysis,
- emergency vehicle identification,
- low-latency local decision making,
- coordinated signal priority across intersections,
- and recovery of normal traffic after preemption.

Recent research has shown that queue spillbacks and recovery of non-priority traffic remain important issues in emergency vehicle preemption [5]. Thus, a system that integrates local AI-based perception with emergency priority and traffic-recovery mechanisms represents a meaningful direction for further investigation.

IV. PROPOSED SYSTEM

The proposed system consists of four main components:

- 1) Street cameras.
- 2) Local smart-pole or edge computers.
- 3) Traffic signal controllers.
- 4) Optional cloud server.

Street cameras capture traffic conditions. The local computer processes the camera information and estimates vehicle conditions. If an emergency vehicle is detected, the local controller can initiate a priority decision.

The cloud server can be used for long-term storage, monitoring, and traffic analysis, but the basic emergency decision does not need to depend entirely on cloud processing.

A. System Architecture

The conceptual architecture is:

**Street Cameras → Local AI Computer → Signal
Controller → Traffic Lights**

An optional cloud connection can be used for data storage and wider traffic analysis.

B. System Flowchart

Fig. 1 shows the proposed decision process.

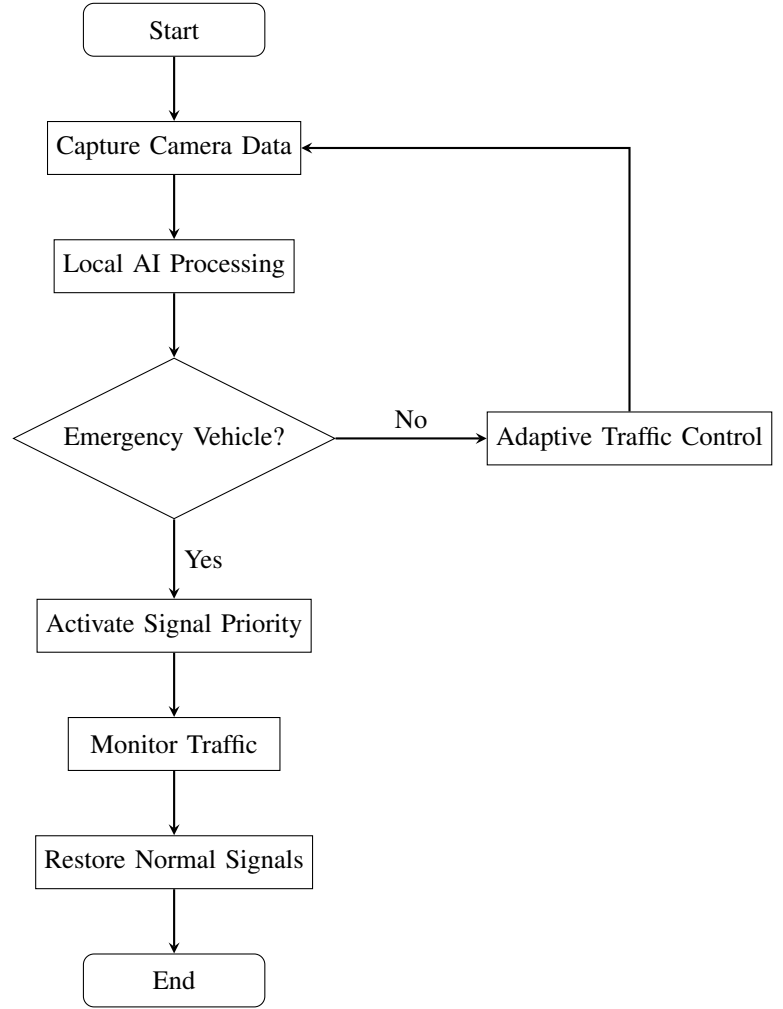


Fig. 1. Flowchart of the proposed local-AI-assisted traffic signal system.

V. MATHEMATICAL MODEL

Let the traffic state at time t be represented by

$$X_t = [x_1, x_2, x_3, \dots, x_n]^T \quad (1)$$

where x_i represents a traffic feature such as vehicle count, queue length, vehicle speed, or emergency-vehicle presence.

The local controller generates a control decision:

$$u_t = f(X_t) \quad (2)$$

where u_t represents the traffic-signal control action.

For emergency vehicle priority, a simplified decision can be represented as

$$u_t = \begin{cases} u_E, & E_t = 1 \\ u_A, & E_t = 0 \end{cases} \quad (3)$$

where E_t indicates whether an emergency vehicle has been detected, u_E represents emergency-priority control, and u_A represents adaptive normal control.

A. Traffic State Matrix

Traffic information from multiple lanes can be represented by

$$X = \begin{bmatrix} q_1 & v_1 & c_1 \\ q_2 & v_2 & c_2 \\ q_3 & v_3 & c_3 \end{bmatrix} \quad (4)$$

where q is queue length, v is average speed, and c is vehicle count.

VI. SYSTEM IMPLEMENTATION

The system can be implemented using a camera, a local computing unit, and a traffic-signal controller.

The local computing unit processes the camera input and generates a short control command. This architecture reduces the need to transmit continuous video to a remote server.

A. Image Insertion

An actual prototype image can be inserted using the following code:

system-image.png

Fig. 2. Prototype or experimental setup of the proposed traffic signal system.

Replace `system-image.png` with your actual image file.

VII. SYSTEM COMPARISON

Table I compares the conceptual characteristics of conventional, cloud-based, and local traffic-control architectures.

The table presents a conceptual comparison rather than measured experimental performance.

TABLE I
CONCEPTUAL COMPARISON OF TRAFFIC CONTROL APPROACHES

Parameter	Timed	Cloud AI	Local AI
Traffic Adaptation	Low	High	High
Internet Dependence	None	High	Low
Local Processing	No	Limited	Yes
Emergency Priority	Limited	Possible	Proposed
Response	Fixed	Network-dependent	Local

VIII. PERFORMANCE EVALUATION

The proposed system should be evaluated using measurable performance indicators rather than assumed values.

Important metrics include:

- Emergency vehicle travel time.
- Emergency vehicle waiting time.
- Average vehicle delay.
- Queue length.
- Signal response time.
- Communication/data usage.

Emergency vehicle delay can be expressed as

$$D_E = T_{\text{actual}} - T_{\text{free-flow}} \quad (5)$$

where T_{actual} is the measured emergency vehicle travel time and $T_{\text{free-flow}}$ is the corresponding free-flow travel time.

A. Results Table

The following table is a template for experimental results. The values should be replaced with measurements from the actual implementation or simulation.

TABLE II
EXPERIMENTAL EVALUATION TEMPLATE

Metric	Baseline	Proposed	Improvement
EV Travel Time	–	–	–
EV Waiting Time	–	–	–
Queue Length	–	–	–
Average Delay	–	–	–
Response Time	–	–	–

No numerical improvement is claimed here because experimental measurements were not provided in the source material.

IX. DISCUSSION

The proposed architecture combines ideas from adaptive traffic signal control, machine vision, and emergency vehicle preemption.

Machine vision can provide traffic information for adaptive control [2], while emergency preemption can provide priority to emergency vehicles [3].

However, emergency priority should not be considered in isolation. Recent research shows that queue spillbacks and the recovery of normal traffic after preemption are important issues [5].

The proposed architecture therefore includes a monitoring and recovery stage after emergency priority is activated.

The main research contribution proposed in this paper is the integration of local AI-based traffic observation with emergency priority and post-preemption traffic recovery.

X. LIMITATIONS

The proposed paper presents an architecture and research direction rather than a completed field deployment.

The following limitations should therefore be considered:

- Camera performance can be affected by weather, lighting, and occlusion.
- Emergency vehicle detection must be sufficiently reliable to avoid incorrect signal priority.
- Coordination between multiple intersections requires additional route information.
- Real-world testing requires suitable traffic-signal infrastructure and safety validation.
- The computational requirements of local AI depend on the selected hardware and detection model.

XI. FUTURE WORK

Future research can investigate:

- 1) Real-time emergency vehicle detection using computer vision.
- 2) Edge AI models optimized for low-power hardware.
- 3) Coordination of multiple traffic intersections.
- 4) Dynamic route-based emergency signal priority.
- 5) Queue-spillback prediction.
- 6) Automatic recovery of traffic signals after emergency preemption.
- 7) Real-world testing using traffic simulation and controlled road experiments.

XII. CONCLUSION

This paper presented a local-AI-assisted smart traffic signal architecture for adaptive traffic management and emergency vehicle priority.

Existing literature demonstrates substantial research in AI-based traffic signal control, machine-vision-based traffic estimation, and emergency vehicle preemption. Therefore, the research gap is more appropriately defined as an integration challenge rather than a complete absence of emergency vehicle priority systems.

The proposed architecture combines camera-based traffic observation, local processing, emergency vehicle detection, signal priority, and post-preemption recovery.

Further experimental work is required to measure response time, emergency vehicle travel time, queue length, and the effect on non-priority traffic.

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